Confining potentials reduce the exotic matter of rotating Ellis–Bronnikov wormholes

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Adding a confining potential to the phantom scalar field significantly reduces the exotic matter required to support a rotating Ellis–Bronnikov wormhole. In the classical massless model, the phantom field is unpinned: it leaks out to spatial infinity with a slow $1/r$ tail and leaves a flat scaling direction that drives radial collapse. Introducing a double-well potential $\mathcal{V}(\psi)$ between the asymptotic vacua creates a localized barrier at the throat that traps the scalar gradient where it supports the geometry and provides a restoring force against radial perturbations. We present an automated search over a four-parameter confining potential at fixed equatorial throat radius R_e and angular momentum J. An evolution strategy (CMA-ES) proposes the potential parameters, and a spectral Newton solver with automatic-differentiation Jacobians constructs the exact stationary metric, verified against unimposed Einstein identities and finer-grid re-solves. Across both symmetric and asymmetric (drainhole) families, trapping the field reduces the volume-integral exoticity quantifier X by 25% to 61% relative to the massless phantom of the same throat size and spin, eliminates the static radial instability, and yields drainhole solutions that hold their throat in three-resolution 3+1 nonlinear evolutions.

I. INTRODUCTION

A static traversable wormhole requires matter that violates the null energy condition (NEC) [3, 4]. The Ellis–Bronnikov throat is the textbook example [1, 2]. It is held open by a single massless phantom scalar ψ with a reversed kinetic term, and it is radially unstable [5, 6, 14, 15]. How little exotic matter a throat needs is a long-standing question. Visser, Kar, and Dadhich showed that the volume-integral quantifier can be made arbitrarily small by concentrating the NEC violation into an infinitesimal thin shell [7]. That surgical construction is not a smooth field theory and cannot be evolved dynamically [8]. Rotation provides a smooth field-theoretic lever: along the spinning families of Kleihaus and Kunz [10, 11], the exotic-matter integral per throat radius falls a hundredfold from the static limit to the fastest spin [15].

However, those spinning families still rely on a free, massless phantom ($\mathcal{V} = 0$). Choosing $\mathcal{V} = 0$ is a matter of analytic convenience rather than physical necessity: without a potential, the scalar equation decouples, but the field itself is completely unpinned. Its gradient decays slowly as $1/r^2$ into the asymptotic regions, wasting exotic energy far from the throat where it does no work supporting the geometry, and the absence of a potential barrier leaves a soft radial mode that drives collapse. This paper asks whether adding a confining potential $\mathcal{V}(\psi)$ to the same phantom field can trap the exotic matter at the throat, reducing the required NEC violation further at fixed throat size and spin while stiffening the throat against radial instability.

The standard route in the wormhole minimization literature is geometry-first: one prescribes a metric or shape function by hand and reads off $T_{\mu\nu} = G_{\mu\nu}/8\pi$ [9, 16]. Most such metrics have no consistent field-theoretic matter realization, and none come with nonlinear 3+1 dynamics. We reverse that order (Fig. 1). The optimizer shapes a confining potential $\mathcal{V}(\psi)$ for the phantom scalar ψ. For each proposed potential, a spectral Newton solver constructs the exact stationary rotating spacetime at fixed equatorial throat radius R_e and angular momentum J. Two Einstein identities that the solver never uses, along with the spectral decay of the highest Chebyshev coefficients, gate every candidate. To our knowledge, this is the first automated minimization of exotic matter over exact, rotating, field-theoretic wormhole solutions.

Section II defines the matter model, the confining potential family, and the exotic-matter integral X. Section III describes the search pipeline: the spectral Newton solver with JAX automatic-differentiation Jacobians [18] and the covariance-matrix adaptation evolution strategy (CMA-ES) [17]. Section IV details the interpolation onto the adaptive-mesh-refinement (AMR) grid, the 3+1 evolution scheme, and the acceptance criteria. Section V reports the optimized minima, their radial stability, and the 3+1 evolution of the slowly rotating drainhole. Section VI summarizes our conclusions. Appendix A records the $\mathcal{V} = 0$ regression and the radial growth-rate screen, and Appendix B presents the three-resolution convergence study of the D21 evolution. We use $G = c = 1$. Following Ref. [15], the drainhole family has throat parameter $\eta_0 = 2$ and end-mass asymmetry $\gamma = -\pi$, so the static member has unit mass.

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II. THE MATTER MODEL

We consider Einstein gravity coupled to the classical Ellis–Bronnikov phantom scalar ψ, augmented by a potential $\mathcal{V}(\psi)$,

$$S = \frac{1}{16\pi}\int d^4x\,\sqrt{-g}\,(R + 2\,g^{\mu\nu}\partial_\mu\psi\,\partial_\nu\psi + 2\,\mathcal{V}(\psi))\,, \quad (1)$$

which is the action of Refs. [10, 11, 15] with $\mathcal{V}(\psi)$ turned on. The reversed kinetic sign allows the stress-energy tensor to violate the NEC, as required to flare out a throat [3]. The potential itself does not alter that sign: $\mathcal{V}(\psi)$ cancels identically between the energy density and the radial pressure, so the NEC violation still tracks the spatial gradient of ψ. Varying Eq. (1) with respect to the metric and ψ gives the coupled Einstein–Klein–Gordon system,

$$R_{\mu\nu} = -2\,\partial_\mu\psi\,\partial_\nu\psi - \mathcal{V}\,g_{\mu\nu}, \qquad \Box\psi = \tfrac{1}{2}\mathcal{V}'(\psi). \quad (2)$$

At $\mathcal{V} = 0$, Eqs. (2) reduce to Eqs. (2)–(3) of Ref. [11], where ψ takes a closed-form arctan profile. When $\mathcal{V} \neq 0$, the potential sources $R^t{}_t + R^\varphi{}_\varphi$, which the three-function Kleihaus–Kunz metric ansatz cannot accommodate. A fourth metric function $B(\eta,\theta)$ therefore joins the line element,

$$\begin{aligned} ds^2 = -e^f dt^2 + B^2 e^{-f}\Big[e^\nu\left(d\eta^2 + h\,d\theta^2\right) \\ + h\sin^2\theta\,(d\varphi - \omega\,dt)^2\Big], \end{aligned} \quad (3)$$

with $h = \eta^2 + \eta_0^2$ and $B \equiv 1$ when $\mathcal{V} = 0$. The five unknowns f, ω, ν, B, and ψ are functions of the radial coordinate η and polar angle θ, while the asymptotic value $\psi(-\infty) = \psi_-$ enters as a bordered unknown determined by the solver. Asymptotically, $f \to 0$ at $\eta \to +\infty$ and $f \to \gamma$ at $\eta \to -\infty$, where $\gamma = 0$ defines the symmetric family of Refs. [10, 11] and $\gamma = -\pi$ defines the asymmetric drainhole family of Ref. [15].

Asymptotically flat ends require degenerate vacua, $\mathcal{V} = \mathcal{V}' = 0$, at both asymptotic field values, $\psi(+\infty) = 0$ and $\psi(-\infty) = \psi_-$. Because $\psi(\eta)$ transitions monotonically from 0 to ψ_- across the wormhole, a double-well potential interpolating between these two vacua vanishes in both asymptotic regions and forms a localized spatial potential barrier right at the throat ($\eta \approx 0$). We parameterize this confining barrier as a topological kink potential,

$$\mathcal{V}(\psi) = \frac{m^2}{\psi_-^2}\,\psi^2(\psi - \psi_-)^2\,Q(\psi), \quad (4)$$

with $Q(u) = 1 + q_1 u + q_2 u^2 + q_3 u^3$ and normalized field coordinate $u = \psi/\psi_- \in [0,1]$ between the vacua.

The four parameters (m, q_1, q_2, q_3) have direct physical interpretations as properties of the throat trap: m sets the scalar mass $\mathcal{V}''(0) = 2m^2$ and thus the Compton confinement scale on the $\eta \to +\infty$ side; $m_- = m\sqrt{1 + q_1 + q_2 + q_3}$ sets the confinement scale on the $\eta \to -\infty$ side; and the relative weights of (q_1, q_2, q_3) sculpt the height and skewness of the potential barrier inside the throat. Requiring $Q(u) > 0$ between the vacua guarantees a positive barrier and preserves the minimum at ψ_-. Setting $m = 0$ recovers the unpinned massless phantom of Ref. [15].

Our objective is the exotic-matter bill of Ref. [15], defined as the Visser–Kar–Dadhich volume-integral quantifier [7] normalized by the equatorial throat radius R_e,

$$X = \frac{1}{R_e}\left|\int(\rho + p_\eta)\,dV\right|. \quad (5)$$

Here ρ is the energy density and p_η is the radial pressure. Although $\mathcal{V}(\psi)$ cancels out of the integrand combination $\rho + p_\eta$, it reduces X indirectly in two ways: the mass and barrier terms compress the gradient $\partial_\eta\psi$ into a compact shell at the throat (eliminating the wasteful $1/r$ tail), and the resulting potential energy depresses the metric factor B in the throat region, lowering the proper volume weight of the NEC violation. If R_e and J are not held fixed, an optimizer can trivially reduce X by shrinking the throat toward an extremal black hole. We therefore impose two additional constraint rows in the solver to pin R_e and J to the exact values of the corresponding $\mathcal{V} = 0$ reference wormhole.

III. OPTIMIZATION

The inner loop (the spectral solver) constructs an exact stationary wormhole for a given potential; the outer loop (CMA-ES) varies the four confining parameters (m, q_1, q_2, q_3) of Eq. (4) to minimize X at fixed R_e and J (Fig. 1).

A. Spectral Newton solver

The compactified coordinate $x = (2/\pi)\arctan(\eta/\eta_0)$ maps both asymptotic ends to $x = \pm 1$. The five fields are expanded in Chebyshev polynomials of x and even cosine modes of θ on a tensor-product Lobatto grid. Damped Newton iteration solves the five coupled elliptic equations for f, ω, $N = \nu/\sin^2\theta$, B, and ψ, bordered by the vacuum value ψ_-, a radial length scale σ, and two constraint rows that pin J and R_e, until the residual reaches machine round-off. Here N is the axis-regularized combination introduced to remove the coordinate degeneracy of the ν equation on the symmetry axis. The dense Jacobian is computed exactly via forward-mode automatic differentiation in `JAX` [18] in column blocks and factored on the GPU via dense LU decomposition. Every converged solution is checked against two first-order Einstein constraint identities that never enter the Newton iteration, as well as the spectral decay of the trailing Chebyshev coefficients. At $\mathcal{V} = 0$, the solver reproduces the published solutions of Ref. [15] to machine precision (Appendix A).

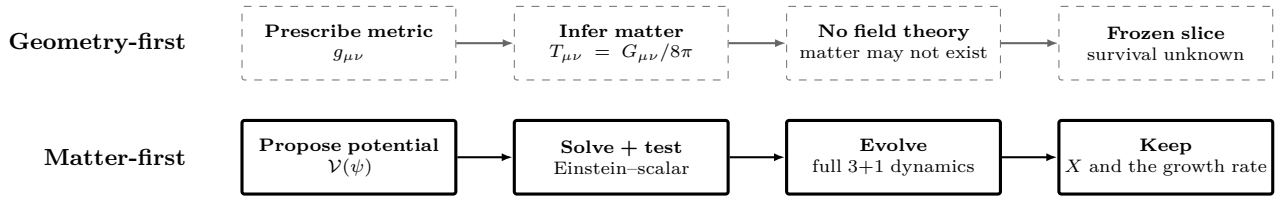


FIG. 1. Geometry-first versus matter-first search. Dashed: prescribe a metric and infer its stress-energy tensor $T_{\mu\nu}$. Solid: propose a confining potential $\mathcal{V}(\psi)$ for the phantom field, solve the coupled Einstein–Klein–Gordon equations at fixed R_e and J, screen for stability, and verify the winners in full 3+1 nonlinear evolution.

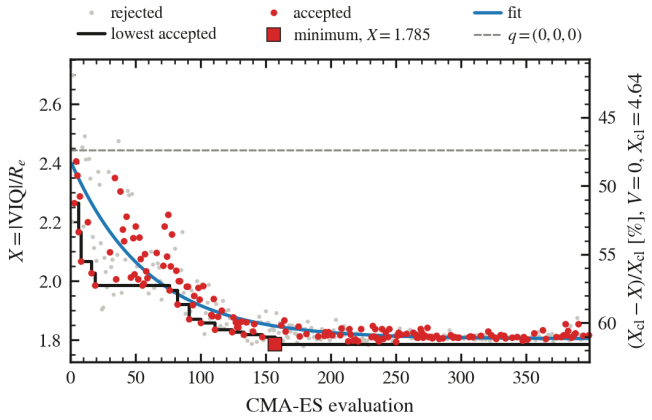


FIG. 2. Exotic-matter bill X across CMA-ES evaluations for the D21 drainhole on the search grid. Red: accepted candidates passing the identity gates; grey: rejected candidates. Dashed line: massless phantom baseline ($\mathcal{V} = 0$, $X = 4.645$). The search-grid minimum ($X = 1.785$) settles at $X = 1.829$ (-61%) upon finer-grid re-solve (Table II).

B. CMA-ES

The outer optimization uses CMA-ES [17] over the four potential parameters (m, q_1, q_2, q_3). Each evaluation performs a full Newton solve, seeded from the nearest previously solved potential in parameter space. Candidates that violate positivity ($Q \leq 0$), fail to converge, or exceed the unimposed Einstein identity thresholds are penalized and rejected rather than ranked as valid minima. To guard against trading exotic matter for fragility, each candidate's static ($J = 0$) counterpart at the same pinned R_e is screened for linear radial ($m = 0$) instabilities, with any positive growth rate penalized in the objective. Every new best clean candidate is re-solved on a finer spectral grid before acceptance. Figure 2 illustrates the CMA-ES search trajectory for the D21 drainhole.

IV. NUMERICAL SETUP

A. Interpolation onto the grid

To evolve the optimized solutions in 3+1 dimensions, the stationary metric (3) is exported as an extended seven-column background table (adding B and the polynomial coefficients of $\mathcal{V}(\psi)$ in the header) and interpolated via C^2 cubic splines onto the Cartesian AMR grid through the isotropic radius r, where $\eta = r - \eta_0^2/4r$ as in Ref. [15]. The $\eta \to -\infty$ asymptotic end compactifies to a puncture at $r = 0$. The conformal factor incorporates B via $\chi \to \chi/B^2$, while the lapse α, shift β^i, and scalar field ψ are read directly from the table. The extrinsic curvature K_{ij} follows analytically from the stationary shift, and the scalar's initial conjugate momentum vanishes in the corotating frame.

B. Evolution scheme

Nonlinear 3+1 evolutions are performed with `GRTeclyn` [22–24] using the numerical setup of Refs. [14, 15]: the conformal and covariant Z4 (CCZ4) formulation [19], 1+log slicing [20], Gamma-driver shift [21], fourth-order spatial finite differences, fourth-order Runge–Kutta time integration with Courant factor $\Delta t = 0.02\,\Delta x$, and Kreiss–Oliger dissipation. The exotic scalar module evaluates the polynomial potential $\mathcal{V}(\psi)$ and its derivative $\mathcal{V}'(\psi)$ directly in the matter evolution kernels and in the Hamiltonian constraint diagnostic. Three nested AMR refinement levels are centered on the throat.

C. Criteria

A candidate solution qualifies as a validated minimum of X only if four gates pass: (i) the Newton residual reaches round-off; (ii) the two unimposed Einstein identities remain below the acceptance threshold in the throat region; (iii) X is invariant under re-solving on a finer spectral grid; and (iv) the two-grid matched $J = 0$ radial perturbation screen finds no unstable mode (Appendix A).

TABLE I. Optimized wormhole models: family, equatorial rotation velocity v_e, and the four confining potential parameters of Eq. (4). Family (a) is the asymmetric drainhole ($\gamma = -\pi$), and family (b) is the symmetric wormhole ($\gamma = 0$). Model Dn has $v_e \approx n/100$; Sn likewise.

model	family	v_e	(m, q_1, q_2, q_3)
S27	(b)	0.27	(0.855, 3.23, 3.78, −0.256)
S50	(b)	0.50	(0.864, 4.61, 0.559, −0.806)
S85	(b)	0.85	(0.425, 2.44, 4.27, 0.445)
D21	(a)	0.21	(1.023, 0.672, 0.048, 0.310)

TABLE II. Exotic-matter integral X of the potential-confined models of Table I, compared against the massless phantom ($\mathcal{V} = 0$) at identical throat radius R_e and angular momentum J. Every model passes the four criteria of Sec. IV C.

model	X	$X_{\mathcal{V}=0}$	ΔX
S27	1.634	2.994	−45%
S50	1.495	2.625	−43%
S85	1.098	1.462	−25%
D21	1.829	4.645	−61%

V. RESULTS

A. Reduction of X at fixed throat and spin

We ran the CMA-ES search over (m, q_1, q_2, q_3) on the symmetric family ($\gamma = 0$) at three rotation rates ($v_e = 0.27, 0.50, 0.85$) and on the asymmetric drainhole family ($\gamma = -\pi$) at $v_e = 0.21$ (Table I). Every reported minimum was verified by re-solving on a finer grid. Table II compares the resulting exotic-matter integral X against the classical massless phantom ($\mathcal{V} = 0$) of the exact same throat radius R_e and spin J.

Confining the phantom field at the throat substantially reduces the exoticity bill at every spin tested: by 45%, 43%, and 25% along the symmetric sequence S27–S85, and by 61% on the slowly rotating drainhole D21. Across all winners, the optimizer selects $m > 0$ and positive barrier shaping ($q_1, q_2 > 0$), steepening the potential walls to localize $\partial_\eta \psi$ near the throat and depressing B below unity in the interior while leavingboth asymptotic ends flat ($B \to 1$). The relative reduction is largest at slow and moderate spins, where the unpinned $\mathcal{V} = 0$ field is most diffuse, and complements the rotational reduction at high spin.

B. Elimination of the radial instability

In thin-shell and ansatz models, squeezing the exotic region often exacerbates radial instabilities. For a field-theoretic confining potential, the opposite occurs. Whereas the massless ($\mathcal{V} = 0$) static symmetric and drainhole backgrounds exhibit the well-known radial collapse modes ($|\omega_I|\eta_0 = 1.182$ and 0.390, respectively), the two-grid matched $J = 0$ radial stability screen finds **no unstable radial mode** on any of the potential-confined backgrounds in Table I (Appendix A). Physically, pinning the scalar field between the two vacua of Eq. (4) removes the flat scaling direction of the massless phantom and supplies a restoring force against radial compression.

C. Three-dimensional evolution of D21

To verify that the potential-confined solution holds dynamically in the full nonlinear regime, we evolved the largest-reduction winner, D21 ($\Delta X = -61\%$), in three spatial dimensions at three resolutions, $N = 96, 128, 192$ (Appendix B and Fig. 3). Across all refinement levels, the Hamiltonian and momentum constraints converge cleanly between second and fourth order, the puncture lapse remains smooth, and both the equatorial and areal throat radii hold steady over the evolution window with relative drift below 10^{-3} at the finest resolution.

VI. CONCLUSIONS

We have demonstrated that trapping the Ellis–Bronnikov phantom scalar with a sculpted confining potential $\mathcal{V}(\psi)$ substantially reduces the exotic matter required to hold a rotating wormhole open at fixed throat size and spin. By coupling a GPU-accelerated spectral Newton solver to a CMA-ES optimizer, our pipeline searches strictly through exact solutions of the Einstein–Klein–Gordon equations gated by unimposed constraint identities. Across symmetric and drainhole families, confining the field to the throat cuts the exotic-matter bill by 25% to 61% relative to the classical massless phantom. Crucially, the potential barrier eliminates the classical $J = 0$ radial instability rather than triggering it, and three-resolution 3+1 evolutions of the 61%-reduced D21 drainhole confirm fourth-order constraint convergence and throat survival in the nonlinear regime.

VII. REPRODUCIBILITY

The spectral Newton solver, the CMA-ES optimization pipeline, the radial stability screen, and the figure generation scripts are openly available in the `GRTeclyn` research repository [25]. Each reported minimum is archived as a seven-column background table with its potential coefficients recorded in the header, serving directly as the input for the 3+1 interpolation of Sec. IV A.

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simulation software. The authors acknowledge the GRTL Collaboration for `GRChombo`, `GRTresna`, and `GRTeclyn`.

Appendix A: Solver and stability checks

At $\mathcal{V} = 0$, the five-field spectral residual reduces symbolically to the published spinning wormhole system of Ref. [15], and Newton iteration across the saved solutions of that work reproduces their global charges, $B \equiv 1$, and Fig. 3e exoticity values X to machine precision. During the $\mathcal{V} \neq 0$ search, the two first-order Einstein equations omitted from the five-equation elliptic solve—the $\eta\theta$ Einstein component and the pointwise generalization of the D^2 conservation identity of Ref. [11]—serve as independent monitors. A candidate is accepted only when both monitors remain below threshold across the throat region and X is confirmed on a finer collocation grid.

The linear radial stability screen computes the spherical ($m = 0$) perturbation spectrum of the $J = 0$ background at the candidate's pinned throat radius R_e in a regular conformal chart. Linearizing the Einstein–Klein–Gordon system yields three coupled radial wave equations cast as a dense generalized eigenvalue problem on Chebyshev–Lobatto grids. Because continuous gauge modes survive discretization at the boundaries, physical eigenvalues are isolated by requiring two-grid eigenvalue convergence alongside two unimposed linearized Einstein constraints (tt and $t\eta$). On the classical $\mathcal{V} = 0$ backgrounds, this pipeline reproduces the published growth rates $|\omega_I|\eta_0 = 0.390$ for the static drainhole [12, 13] and $|\omega_I|\eta_0 = 1.182$ for the symmetric Ellis throat to four significant digits. Applied to the potential-confined models of Table I, the screen finds no unstable radial mode up to $N_x = 300$.

Appendix B: Convergence of the D21 evolution

The optimized D21 drainhole of Table I was evolved in `GRTeclyn` at three base resolutions, $N = 96, 128, 192$, on a domain of length $L = 64$ with three nested refinement levels. Because the exact continuum values of the Hamiltonian and momentum constraints vanish, rescaling the L_2 constraint norms directly measures the convergence order of the coupled metric–potential implementation. Figure 3 displays the Hamiltonian and momentum constraint norms on each refinement level alongside the $N = 192$ curves rescaled by 2^2 (second order) and 2^4 (fourth order), as well as the equatorial throat radius $R_{\rm eq}$, areal radius R, gravitational-wave mode Ψ_4^{20} at $R = 14$, and minimum puncture lapse $\alpha_{\rm min}$. The constraints converge cleanly toward zero with increasing resolution, and the throat geometry remains stable throughout the evolution.

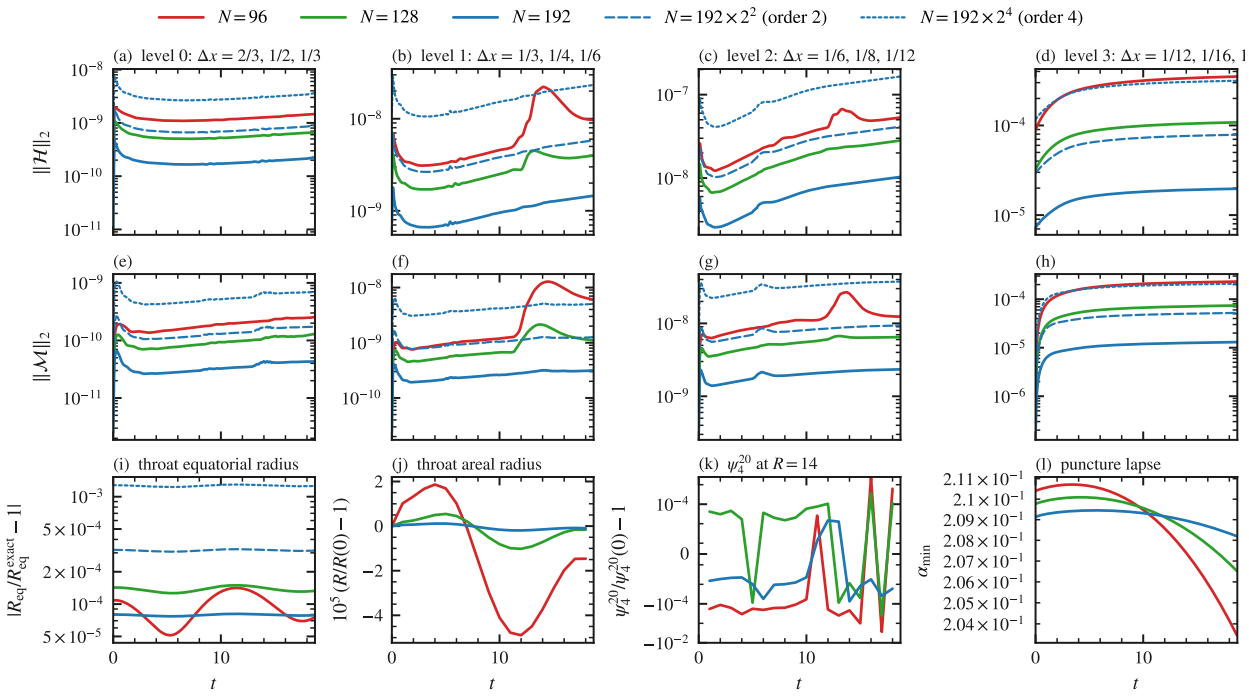


FIG. 3. Three-resolution convergence study of the optimized D21 drainhole evolution over $N = 96, 128, 192$. (a)–(d) Hamiltonian and (e)–(h) momentum constraint L_2 norms per refinement level; dashed and dotted curves show the $N = 192$ norm rescaled for second-order (2^2) and fourth-order (2^4) convergence. (i)–(l) Equatorial throat radius drift, areal throat radius drift, $(\ell, m) = (2, 0)$ Newman–Penrose scalar Ψ_4^{20} extracted at $R = 14$, and minimum puncture lapse $\alpha_{\min}$. Constraint norms fall systematically with resolution and the throat holds.

[1] H. G. Ellis, "Ether flow through a drainhole: A particle model in general relativity," J. Math. Phys. **14**, 104 (1973).

[2] K. A. Bronnikov, "Scalar-tensor theory and scalar charge," Acta Phys. Polon. B **4**, 251 (1973).

[3] M. S. Morris and K. S. Thorne, "Wormholes in spacetime and their use for interstellar travel: A tool for teaching general relativity," Am. J. Phys. **56**, 395 (1988).

[4] M. Visser, *Lorentzian Wormholes: From Einstein to Hawking* (AIP Press, Woodbury, NY, 1995).

[5] J. A. González, F. S. Guzmán, and O. Sarbach, "Instability of wormholes supported by a ghost scalar field. I. Linear stability analysis," Class. Quantum Grav. **26**, 015010 (2009).

[6] J. A. González, F. S. Guzmán, and O. Sarbach, "Instability of wormholes supported by a ghost scalar field. II. Nonlinear evolution," Class. Quantum Grav. **26**, 015011 (2009).

[7] M. Visser, S. Kar, and N. Dadhich, "Traversable wormholes with arbitrarily small energy condition violations," Phys. Rev. Lett. **90**, 201102 (2003).

[8] C. J. Fewster and T. A. Roman, "On wormholes with arbitrarily small quantities of exotic matter," Phys. Rev. D **72**, 044023 (2005).

[9] P. K. F. Kuhfittig, "Viable models of traversable wormholes supported by small amounts of exotic matter," arXiv:0806.1194 [gr-qc] (2008).

[10] B. Kleihaus and J. Kunz, "Rotating Ellis wormholes in four dimensions," Phys. Rev. D **90**, 121503(R) (2014).

[11] X. Y. Chew, B. Kleihaus, and J. Kunz, "Geometry of spinning Ellis wormholes," Phys. Rev. D **94**, 104031 (2016).

[12] B. Azad, J. L. Blázquez-Salcedo, F. S. Khoo, and J. Kunz, "Are slowly rotating Ellis–Bronnikov wormholes stable?," Phys. Lett. B **848**, 138349 (2024).

[13] B. Azad, J. L. Blázquez-Salcedo, F. S. Khoo, and J. Kunz, "Radial perturbations of Ellis–Bronnikov wormholes in slow rotation up to second order," Phys. Rev. D **109**, 124051 (2024).

[14] N. M. Shirokov, "Wormhole dynamics: nonlinear collapse and gravitational-wave emission," arXiv:2604.00071 [gr-qc] (2026).

[15] N. M. Shirokov and I. Nachevsky, "Rotation slows the instability of Ellis–Bronnikov wormholes," arXiv:2610.29941 [gr-qc] (2026).

[16] C. Helmerich, J. Fuchs, A. Bobrick, L. Sellers, B. Melcher, and G. Martire, "Analyzing warp drive spacetimes with Warp Factory," Class. Quantum Grav. **41**, 095009 (2024).

[17] N. Hansen and A. Ostermeier, "Completely derandomized self-adaptation in evolution strategies," Evol. Comput. **9**, 159 (2001).

[18] J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Necula, A. Paszke, J. VanderPlas, S. Wanderman-Milne, and Q. Zhang, `JAX`: composable transformations of Python+NumPy programs, https://github.com/jax-ml/jax (2018).

[19] D. Alic, C. Bona-Casas, C. Bona, L. Rezzolla, and C. Palenzuela, "Conformal and covariant formulation of the Z4 system with constraint-violation damping," Phys. Rev. D **85**, 064040 (2012).

[20] C. Bona, J. Massó, E. Seidel, and J. Stela, "New formalism for numerical relativity," Phys. Rev. Lett. **75**, 600 (1995).

[21] M. Alcubierre, B. Brügmann, P. Diener, M. Koppitz, D. Pollney, E. Seidel, and T. Takahashi, "Gauge conditions for long-term numerical black hole evolutions without excision," Phys. Rev. D **67**, 084023 (2003).

[22] GRTL Collaboration, `GRTeclyn`, https://github.com/GRTLCollaboration/GRTeclyn.

[23] W. Zhang *et al.*, "AMReX: a framework for block-structured adaptive mesh refinement," J. Open Source Softw. **4**, 1370 (2019).

[24] T. Andrade *et al.*, "GRChombo: an adaptable numerical relativity code for fundamental physics," J. Open Source Softw. **6**, 3703 (2021).

[25] N. M. Shirokov and I. Nachevsky, `GRTeclyn` research fork, https://github.com/Nikchik-coder/GRTeclyn, doi:10.5281/zenodo.23245933 (2026).